

Install Library

```
%pip install mlflow
```

```
Collecting mlflow
```

```
  Downloading mlflow-3.12.0-py3-none-any.whl.metadata (49 kB)
```

4

```
Collecting mlflow-skinny==3.12.0 (from mlflow)
```

```
  Downloading mlflow_skinny-3.12.0-py3-none-any.whl.metadata (50 kB)
```

```
Collecting mlflow-tracing==3.12.0 (from mlflow)
```

```
  Downloading mlflow_tracing-3.12.0-py3-none-any.whl.metadata (19 kB)
```

```
Collecting Flask-CORS<7 (from mlflow)
```

```
  Downloading flask_cors-6.0.2-py3-none-any.whl.metadata (5.3 kB)
```

```
Requirement already satisfied: Flask<4 in /usr/local/lib/python3.12/dist-pa
```

```
Requirement already satisfied: aiohttp<4 in /usr/local/lib/python3.12/dist-
```

```
Requirement already satisfied: alembic!=1.10.0,<2 in /usr/local/lib/python3
```

```
Collecting cryptography<47,>=43.0.0 (from mlflow)
```

```
  Downloading cryptography-46.0.7-cp311-abi3-manylinux_2_34_x86_64.whl.meta
```

```
Collecting docker<8,>=4.0.0 (from mlflow)
```

```
  Downloading docker-7.1.0-py3-none-any.whl.metadata (3.8 kB)
```

```
Collecting graphene<4 (from mlflow)
```

```
  Downloading graphene-3.4.3-py2.py3-none-any.whl.metadata (6.9 kB)
```

```
Collecting gunicorn<26 (from mlflow)
```

```
  Downloading gunicorn-25.3.0-py3-none-any.whl.metadata (5.5 kB)
```

```
Collecting huey<3,>=2.5.4 (from mlflow)
```

```
  Downloading huey-2.6.0-py3-none-any.whl.metadata (4.3 kB)
```

```
Requirement already satisfied: matplotlib<4 in /usr/local/lib/python3.12/di
```

```
Requirement already satisfied: numpy<3 in /usr/local/lib/python3.12/dist-pa
```

```
Requirement already satisfied: pandas<3 in /usr/local/lib/python3.12/dist-p
```

```
Requirement already satisfied: pyarrow<24,>=4.0.0 in /usr/local/lib/python3
```

```
Requirement already satisfied: scikit-learn<2 in /usr/local/lib/python3.12/
```

```
Requirement already satisfied: scipy<2 in /usr/local/lib/python3.12/dist-pa
```

```
Collecting skops<1 (from mlflow)
```

```
  Downloading skops-0.14.0-py3-none-any.whl.metadata (4.4 kB)
```

```
Requirement already satisfied: sqlalchemy<3,>=1.4.0 in /usr/local/lib/pytho
```

```
Requirement already satisfied: cachetools<8,>=5.0.0 in /usr/local/lib/pytho
```

```
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.12/d
```

```
Requirement already satisfied: cloudpickle<4 in /usr/local/lib/python3.12/d
```

```
Collecting databricks-sdk<1,>=0.20.0 (from mlflow-skinny==3.12.0->mlflow)
```

```
  Downloading databricks_sdk-0.112.0-py3-none-any.whl.metadata (43 kB)
```

```
Requirement already satisfied: fastapi<1 in /usr/local/lib/python3.12/dist-
```

```
Requirement already satisfied: gitpython<4,>=3.1.9 in /usr/local/lib/python
```

```
Requirement already satisfied: importlib_metadata!=4.7.0,<10,>=3.7.0 in /us
```

```
Requirement already satisfied: opentelemetry-api<3,>=1.9.0 in /usr/local/li
```

```
Requirement already satisfied: opentelemetry-proto<3,>=1.9.0 in /usr/local/
```

```
Requirement already satisfied: opentelemetry-sdk<3,>=1.9.0 in /usr/local/li
```

```
Requirement already satisfied: packaging<27 in /usr/local/lib/python3.12/di
```

```
Requirement already satisfied: protobuf<8,>=3.12.0 in /usr/local/lib/python
```

```
Requirement already satisfied: pydantic<3,>=2.0.0 in /usr/local/lib/python3
```

```
Requirement already satisfied: python-dotenv<2,>=0.19.0 in /usr/local/lib/p
```

```
Requirement already satisfied: pyyaml<7,>=5.1 in /usr/local/lib/python3.12/
Requirement already satisfied: requests<3,>=2.17.3 in /usr/local/lib/python
Requirement already satisfied: sqlparse<1,>=0.4.0 in /usr/local/lib/python3
Requirement already satisfied: starlette<1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: typing-extensions<5,>=4.0.0 in /usr/local/li
Requirement already satisfied: uvicorn<1 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/py
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.1
```

Import Library

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
import mlflow
import mlflow.sklearn

# Scikit-learn datasets & model selection
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

#Evaluation matrices - THE CORE OF THIS NOTEBOOK
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    ConfusionMatrixDisplay
)
```

Load and Split Data

```
data = load_breast_cancer()
X, y = data.data, data.target
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(X, y, test_size=
print(f"Train: {X_train_c.shape} Test: {X_test_c.shape}")
```

```
Train: (455, 30) Test: (114, 30)
```

Configure MLflow tracking URI and experiment

```
mlflow.set_tracking_uri("sqlite:///mlflow.db")

mlflow.set_experiment("breast-cancer-classifier")
```

```

2026/05/30 11:33:11 INFO mlflow.store.db.utils: Creating initial MLflow data
2026/05/30 11:33:11 INFO mlflow.store.db.utils: Updating database tables
2026/05/30 11:33:14 INFO mlflow.tracking.fluent: Experiment with name 'breast
<Experiment: artifact_location='/content/mlruns/1',
creation_time=1780140794139, experiment_id='1',
last_update_time=1780140794139, lifecycle_stage='active', name='breast-
cancer-classifier', tags={}, trace_location=None, workspace='default'>

```

Manual Log

Start coding or [generate](#) with AI.

```

MAX_DEPTH = 4
MIN_SAMPLES = 5
CRITERION = 'gini'

mlflow.end_run()

with mlflow.start_run(run_name="manual-logging") as run :
    mlflow.log_param("max_depth", MAX_DEPTH)
    mlflow.log_param("min_sample_split", MIN_SAMPLES)
    mlflow.log_param("criterion", CRITERION)
    mlflow.log_param("random_state", 42 )

clf = DecisionTreeClassifier(
    max_depth=MAX_DEPTH,
    min_samples_split=MIN_SAMPLES,
    criterion=CRITERION,
    random_state=42
)
clf.fit(X_train_c, y_train_c)
y_pred = clf.predict(X_test_c)

#log Evaluation Matrics
mlflow.log_metric("accuracy", accuracy_score(y_test_c, y_pred))
mlflow.log_metric("precision", precision_score(y_test_c, y_pred))
mlflow.log_metric("recall", recall_score(y_test_c, y_pred))
mlflow.log_metric("f1", f1_score(y_test_c, y_pred))

#Save confusion matrix as a PNG artifact
fig, ax = plt.subplots(figsize=(5, 4))
ConfusionMatrixDisplay.from_predictions(y_test_c, y_pred, ax=ax)
plt.tight_layout()
plt.savefig("confusion_matrix.png")
mlflow.log_artifact("confusion_matrix.png")
plt.close()

mlflow.sklearn.log_model(clf, "decision-tree-model")

```

```

print(f"RUN ID: {run.info.run_id}")
print(f"Accuracy: {accuracy_score(y_test_c, y_pred):.4f}")

mlflow.sklearn.autolog(
    log_input_examples = True,
    log_model_signatures = True,
    log_models = True
)
mlflow.end_run()
with mlflow.start_run(run_name="autolog-run"):
    clf2 = DecisionTreeClassifier(max_depth=6, random_state=42)
    clf2.fit(X_train_c, y_train_c)
    print("autolog run complete - check the UI for logged param & metrics")

!mlflow ui --backend-store-uri sqlite:///mlflow.db --port 5000 &
print("UI running at: http://localhost:5000")

```

```

2026/05/30 11:57:19 INFO mlflow.utils.autologging_utils: Created MLflow autol
2026/05/30 11:57:20 WARNING mlflow.sklearn: Saving scikit-learn models in the
2026/05/30 11:57:27 WARNING mlflow.models.model: `artifact_path` is deprecate
2026/05/30 11:57:27 WARNING mlflow.sklearn: Saving scikit-learn models in the
2026/05/30 11:57:31 WARNING mlflow.sklearn: Saving scikit-learn models in the
RUN ID: fc5380bd93734a98891aaad2262d5d3e
Accuracy: 0.9386
autolog run complete - check the UI for logged param & metrics
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only).
2026/05/30 11:57:47 INFO: Uvicorn running on http://127.0.0.1:5000 (Press
2026/05/30 11:57:47 INFO: Started parent process [12283]
2026/05/30 11:58:24 INFO: Started server process [12324]
2026/05/30 11:58:24 INFO: Waiting for application startup.
2026/05/30 11:58:24 INFO: Application startup complete.
2026/05/30 11:58:24 INFO: Started server process [12325]
2026/05/30 11:58:24 INFO: Waiting for application startup.
2026/05/30 11:58:24 INFO: Application startup complete.
2026/05/30 11:58:24 INFO: Started server process [12327]
2026/05/30 11:58:24 INFO: Waiting for application startup.
2026/05/30 11:58:24 INFO: Application startup complete.
2026/05/30 11:58:24 INFO: Started server process [12326]
2026/05/30 11:58:24 INFO: Waiting for application startup.
2026/05/30 11:58:24 INFO: Application startup complete.
2026/05/30 11:58:24 INFO mlflow.server.jobs.utils: Registered online_scoring_
2026/05/30 12:11:53 INFO: Shutting down
2026/05/30 12:11:53 INFO: Shutting down
2026/05/30 12:11:53 INFO: Shutting down
2026/05/30 12:11:53 INFO: Shutting down
2026/05/30 12:11:53 INFO: Waiting for application shutdown.
2026/05/30 12:11:53 INFO: Application shutdown complete.
2026/05/30 12:11:53 INFO: Finished server process [12325]
2026/05/30 12:11:53 INFO: Waiting for application shutdown.
2026/05/30 12:11:53 INFO: Application shutdown complete.
2026/05/30 12:11:53 INFO: Finished server process [12327]
2026/05/30 12:11:53 INFO: Waiting for application shutdown.

```

```

2026/05/30 12:11:53 INFO: Application shutdown complete.
2026/05/30 12:11:53 INFO: Finished server process [12324]
2026/05/30 12:11:53 INFO: Waiting for application shutdown.
2026/05/30 12:11:53 INFO: Application shutdown complete.
2026/05/30 12:11:53 INFO: Finished server process [12326]
2026/05/30 12:11:54 INFO: Received SIGINT, exiting.
2026/05/30 12:11:54 INFO: Terminated child process [12324]
2026/05/30 12:11:54 INFO: Terminated child process [12325]
2026/05/30 12:11:54 INFO: Terminated child process [12326]
2026/05/30 12:11:54 INFO: Terminated child process [12327]
2026/05/30 12:11:54 INFO: Waiting for child process [12324]
2026/05/30 12:11:54 INFO: Waiting for child process [12325]
2026/05/30 12:11:54 INFO: Waiting for child process [12326]
2026/05/30 12:11:54 INFO: Waiting for child process [12327]
2026/05/30 12:11:54 INFO: Stopping parent process [12283]
UI running at: http://localhost:5000

```

Student Assignment

```

import mlflow, mlflow.sklearn
from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, f1_score

mlflow.set_tracking_uri("sqlite:///quiz.db")
mlflow.set_experiment("quiz-experiment")

data = load_iris()
X, y = data.data, data.target
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=

MAX_DEPTH = 5
N_ESTIMATORS = 100

with mlflow.start_run(run_name="quiz-run"):
    #1. Log Params
    mlflow.log_param("max_depth", MAX_DEPTH)
    mlflow.log_param("n_estimators", N_ESTIMATORS)

    model = RandomForestClassifier(
        max_depth = MAX_DEPTH, n_estimators=N_ESTIMATORS, random_state=42
    )
    model.fit(X_tr, y_tr)
    preds = model.predict(X_te)

    #2. Log Metrics
    mlflow.log_metric("accuracy", accuracy_score(y_te, preds))
    mlflow.log_metric("f1", f1_score(y_te, preds, average='macro'))

```

```
#3. Log and register model
mlflow.sklearn.log_model(
    model,
    "random-forest-model",
    registered_model_name = "iris-rf-classifier"
)

#4. Set Tag
mlflow.set_tag("team", "data-science")
```

```
2026/05/30 12:11:56 INFO mlflow.store.db.utils: Creating initial MLflow datab
2026/05/30 12:11:56 INFO mlflow.store.db.utils: Updating database tables
2026/05/30 12:11:58 INFO mlflow.tracking.fluent: Experiment with name 'quiz-e
2026/05/30 12:11:59 WARNING mlflow.sklearn: Saving scikit-learn models in the
2026/05/30 12:12:03 WARNING mlflow.models.model: `artifact_path` is deprecate
2026/05/30 12:12:04 WARNING mlflow.sklearn: Saving scikit-learn models in the
Successfully registered model 'iris-rf-classifier'.
Created version '1' of model 'iris-rf-classifier'.
```